

Tribhuvan University

Faculty of Humanities and Social Sciences

Masters of Computer Application (MCA)

1st Semester

Advanced Database Management System(MCA-503)

Unit-6: Big Data Technologies Based on MapReduce and Hadoop 5hrs

Instructor

Tekendra Nath Yogi (M.SC CSIT TU)

Tekendranath@gmail.com

Contents

- **Big Data Technologies Based on MapReduce and Hadoop**
[5 Hrs.]
 - 6.1. Introduction to Big Data;
 - 6.2. Introduction to MapReduce and Hadoop;
 - 6.3. Hadoop Distributed File Systems (HDFS);
 - 6.4. MapReduce: Additional Details;
 - 6.5. Hadoop v2 alias YARN.

Introduction

- Phenomenal growth in data generation
 - Social media
 - Sensors
 - Communications networks and satellite imagery
 - User-specific business data
- “Big data” refers to massive amounts of data
 - Exceeds the typical reach of a DBMS
- Big data ranges from terabytes (10^{12} bytes) or petabytes (10^{15} bytes) to exabytes (10^{18} bytes)

6.1 What is Big Data?

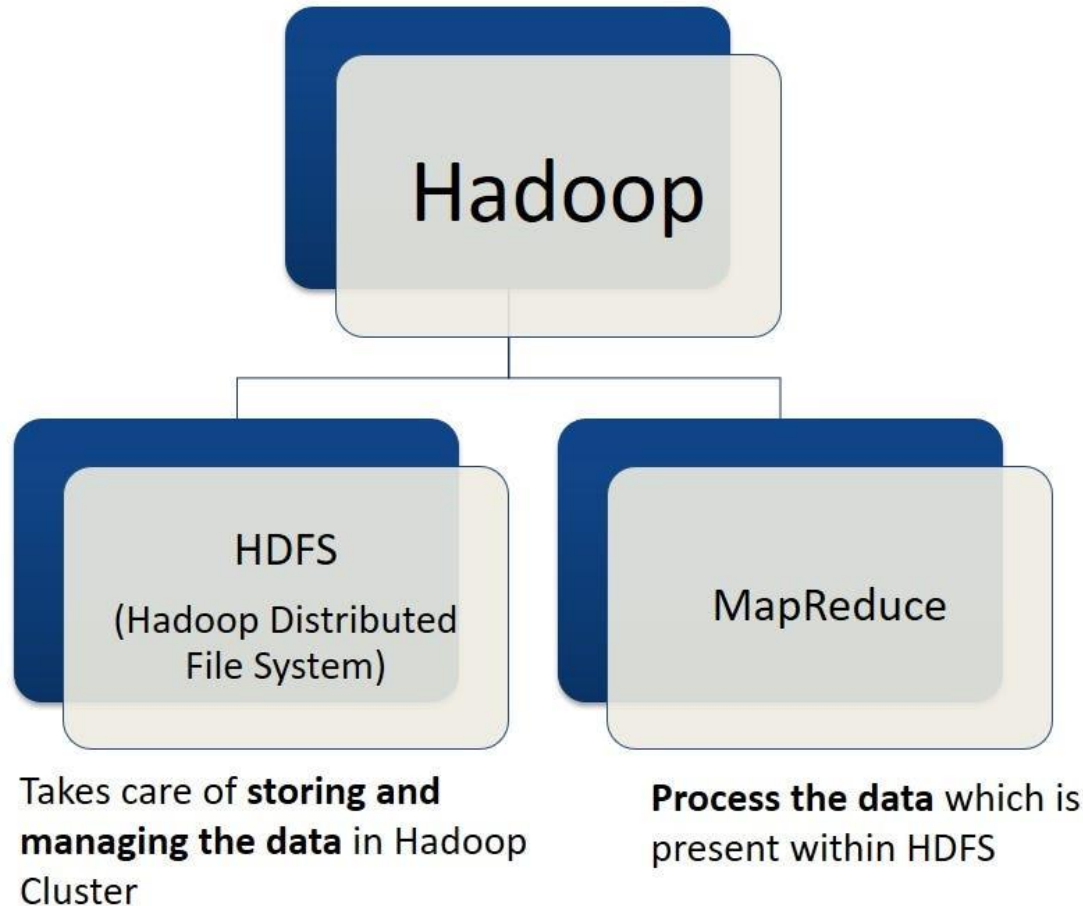


Cont'd...

- Volume
 - Refers to size of data managed by the system
- Velocity
 - Speed of data creation, ingestion, and processing
- Variety
 - Refers to type of data source
 - Structured, unstructured
- Veracity
 - Credibility of the source
 - Suitability of data for the target audience
 - Evaluated through quality testing or credibility analysis
- Value
 - Useful information obtained from data

6.2 Introduction to MapReduce and Hadoop

- Core components of Hadoop are MapReduce programming paradigm and Hadoop Distributed File System (HDFS)



Cont'd...

- Hadoop originated from quest for open source search engine
 - Developed by Cutting and Carafella in 2004
 - Cutting joined Yahoo in 2006
 - Yahoo spun off Hadoop-centered company in 2011
 - Tremendous growth

Cont'd...

- MapReduce
 - Fault-tolerant implementation and runtime environment
 - Developed by Dean and Ghemawat at Google in 2004
 - Programming style: map and reduce tasks
 - Automatically parallelized and executed on large clusters of commodity hardware
 - Allows programmers to analyze very large datasets
 - Underlying data model assumed: key-value pair

The MapReduce Programming Model

- Map
 - Generic function that takes a key of type K1 and value of type V1
 - Returns a list of key-value pairs of type K2 and V2
- Reduce
 - Generic function that takes a key of type K2 and a list of values V2 and returns pairs of type (K3, V3)
- Outputs from the map function must match the input type of the reduce function

Cont'd...

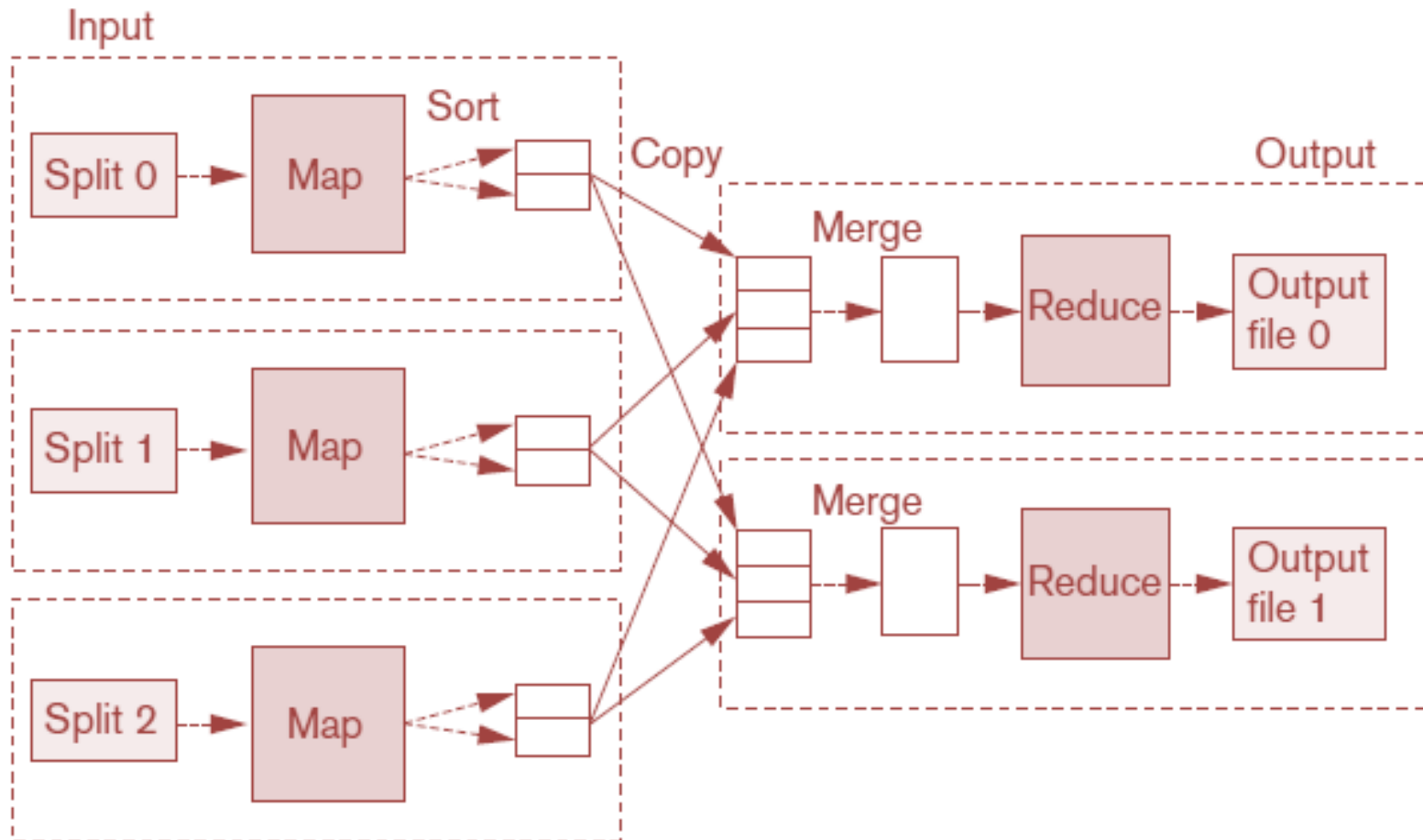


Figure: Overview of MapReduce execution

Cont'd...

- MapReduce example

- Make a list of frequencies of words in a document
- Pseudocode

Map (String key, String value):

for each word w in value Emitintermediate (w, "1");

Reduce (String key, Iterator values) : // here the key is a word and values are lists of its counts //

Int result =0;

For each v in values :

result += Parseint (v);

Emit (key, Asstring (result));

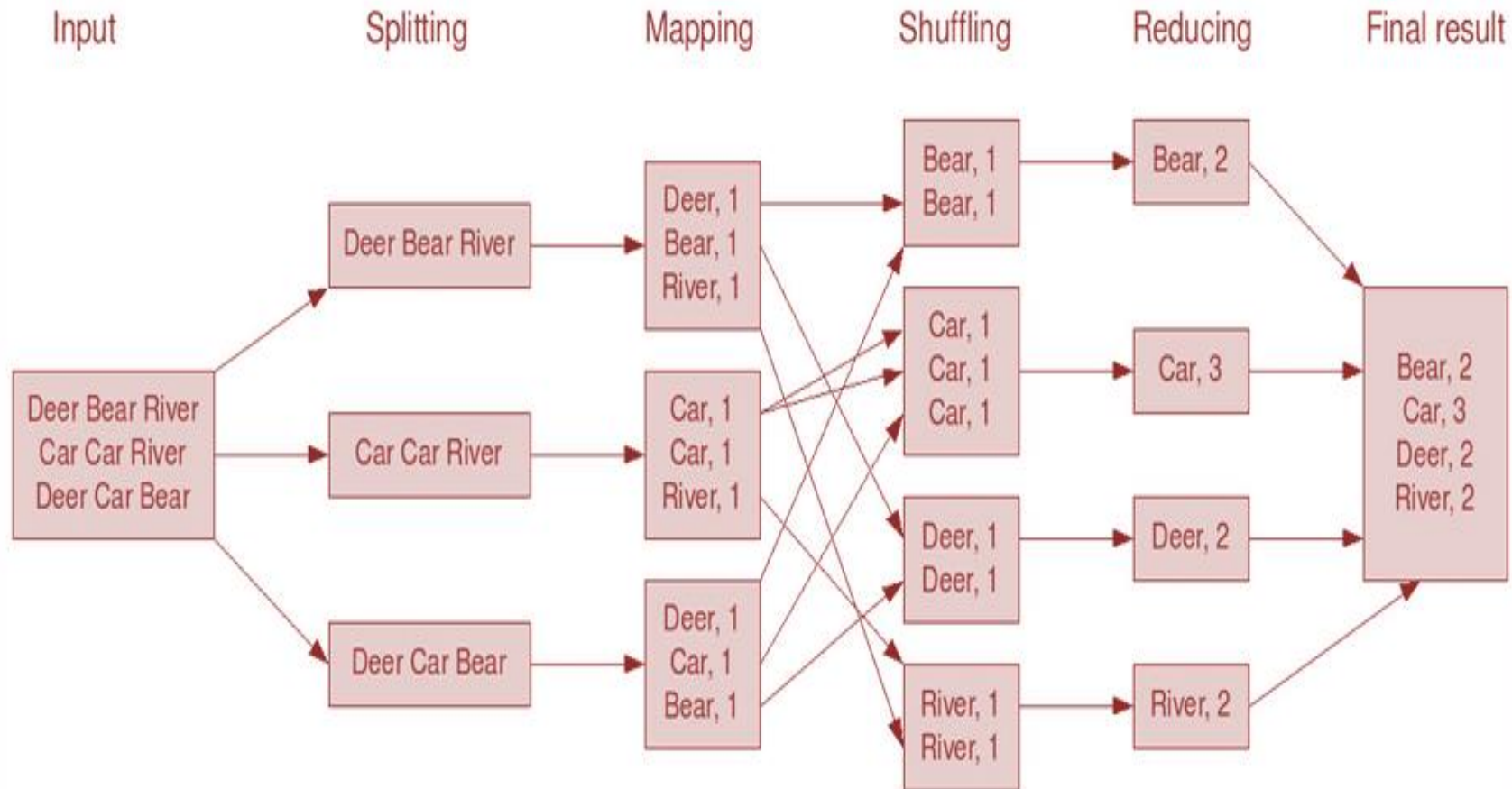
Cont'd...

- MapReduce example (cont'd.): Actual MapReduce code

```
map[LongWritable,Text](key, value) : List[Text, LongWritable] = {  
    String[] words = split(value)  
    for(word : words) {  
        context.out(Text(word), LongWritable(1))  
    }  
}  
reduce[Text, Iterable[LongWritable]](key, values) : List[Text, LongWritable] = {  
    LongWritable c = 0  
    for( v : values) {  
        c += v  
    }  
    context.out(key,c)  
}
```

Cont'd...

- Map Reduce For example, Word Frequency count



Cont'd...

- **Hadoop releases:**



Hadoop v1.0

MapReduce

Data Processing
& Resource Management

HDFS

Distributed File Storage



Hadoop v2.0

MapReduce

**Other Data
Processing
Frameworks**

YARN

Resource Management

HDFS

Distributed File Storage

Cont'd...

- **Hadoop releases: 1.x features**

- Continuation of the original code base
- Additions include security, additional HDFS and MapReduce improvements

- **Hadoop releases: 2.x features**

- YARN (Yet Another Resource Navigator)
- A new MR runtime that runs on top of YARN
- Improved HDFS that supports federation and increased availability

6.3 Hadoop Distributed File System (HDFS)

- HDFS
 - File system component of Hadoop
 - Designed to run on a cluster of commodity hardware
 - Patterned after UNIX file system
 - Provides high-throughput access to large datasets
 - Stores metadata on **NameNode** server
 - Stores application data on **DataNode** servers
 - File content replicated on multiple DataNodes

Cont'd...

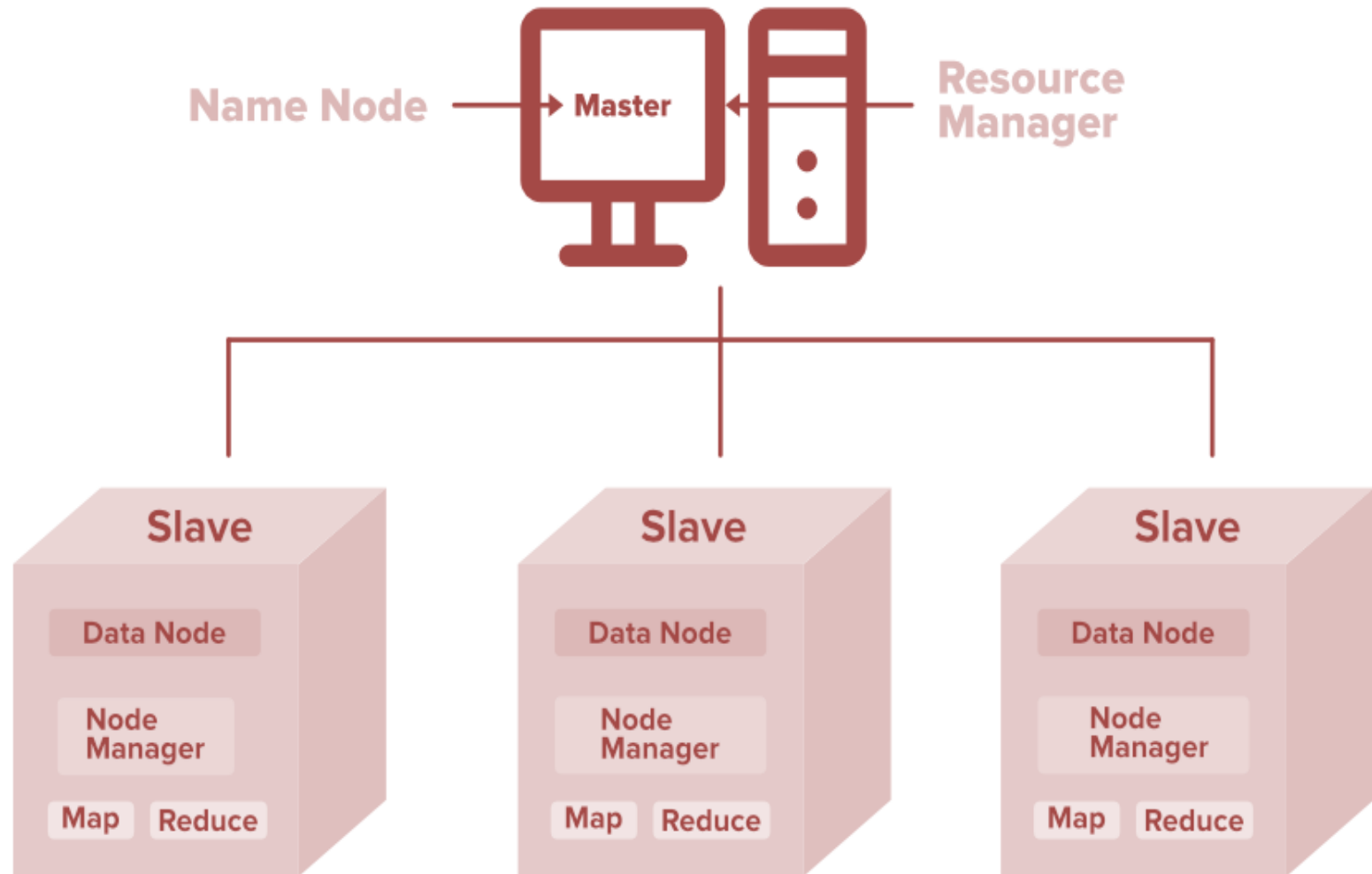
- HDFS design assumptions and goals
 - Hardware failure is the norm or common
 - Batch processing
 - Large datasets
 - Simple coherency model – write once read repeatedly

Cont'd...

- HDFS architecture

- Master-slave: divides workload between the name node and data node.
- Decouples metadata from data operations by using Name node and data node.
- Replication provides reliability and high availability
- Network traffic minimized as it supports the data close computation principle along with write pipeline.

Cont'd...



Cont'd...

- **Name Node:**
 - Maintains image of the file system
 - i-nodes and corresponding block locations
 - Changes maintained in write-ahead commit log called Journal
- **Secondary Name Nodes**
 - Checkpointing role or backup role
- **Data Nodes**
 - Stores blocks in node's native file system
 - Periodically reports state to the Name Node

Cont'd...

- **File I/O operations**

- Single-writer, multiple-reader model
- Files cannot be updated, only appended
- Write pipeline set up to minimize network utilization

- **Block placement**

- Nodes of Hadoop cluster typically spread across many racks
 - Nodes on a rack share a switch

Cont'd...

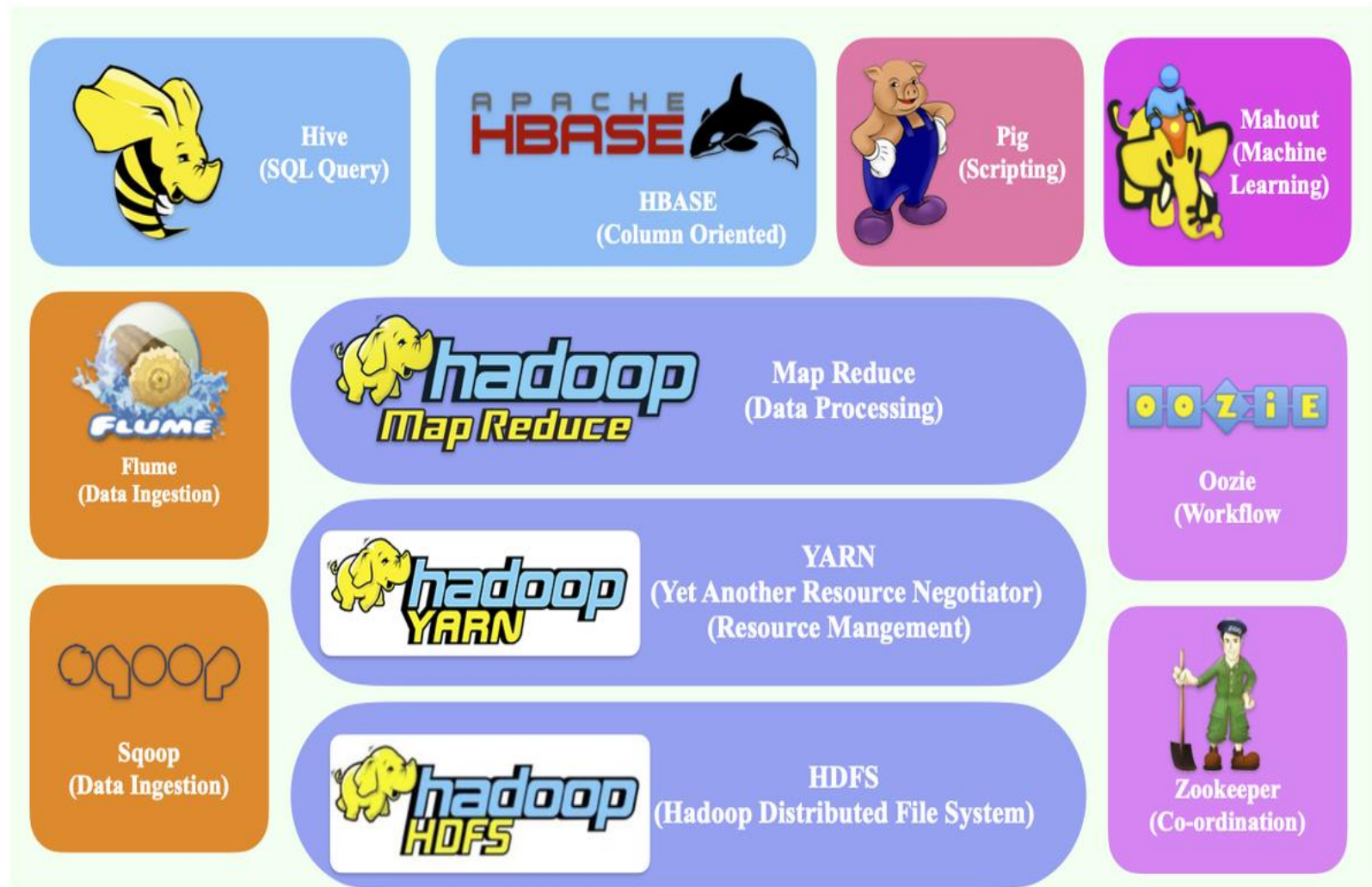
- **Replica management**

- Name Node tracks number of replicas and block location
 - Based on block reports
- Replication priority queue contains blocks that need to be replicated

- **HDFS scalability**

- Yahoo cluster achieved 14 petabytes, 4000 nodes, 15k clients, and 600 million files

The Hadoop Ecosystem



Cont'd...

- Related projects with additional functionality
 - Pig and hive
 - Provides higher-level interface for working with Hadoop framework
 - Oozie
 - Service for scheduling and running workflows of jobs
 - Sqoop
 - Library and runtime environment for efficiently moving data between relational databases and HDFS
 - HBase
 - Column-oriented key-value store that uses HDFS

6.4 MapReduce: Additional Details

- MapReduce runtime environment
 - JobTracker
 - Master process
 - Responsible for managing the life cycle of Jobs and scheduling Tasks on the cluster
 - TaskTracker
 - Slave process
 - Runs on all Worker nodes of the cluster

Cont'd...

- Overall flow of a MapReduce job
 - Job submission
 - Job initialization
 - Task assignment
 - Task execution
 - Job completion

Cont'd...

- Fault tolerance in MapReduce
 - Task failure
 - Runtime exception
 - Java virtual machine crash
 - No timely updates from the task process
 - TaskTracker failure
 - Crash or disconnection from JobTracker
 - Failed Tasks are rescheduled
 - JobTracker failure
 - Not a recoverable failure in Hadoop v1

Cont'd...

- The shuffle procedure
 - Reducers get all the rows for a given key together
 - Map phase
 - Background thread partitions buffered rows based on the number of Reducers in the job and the Partitioner
 - Rows sorted on key values
 - Comparator or Combiner may be used
 - Copy phase
 - Reduce phase

Cont'd...

- Job scheduling
 - Job Tracker schedules work on cluster nodes
 - Fair Scheduler
 - Provides fast response time to small jobs in a Hadoop shared cluster
 - Capacity Scheduler
 - Geared to meet needs of large enterprise customers

Cont'd...

- Strategies for equi-joins in MapReduce environment
 - Sort-merge join
 - Map-side hash join
 - Partition join
 - Bucket joins
 - N-way map-side joins
 - Simple N-way joins

Cont'd...

- Apache Pig
 - Bridges the gap between declarative-style interfaces such as SQL, and rigid style required by MapReduce
 - Designed to solve problems such as ad hoc analyses of Web logs and clickstreams
 - Accommodates user-defined functions

Cont'd...

- Apache Hive
 - Provides a higher-level interface to Hadoop using SQL-like queries
 - Supports processing of aggregate analytical queries typical of data warehouses
 - Developed at Facebook

Hive System Architecture and Components

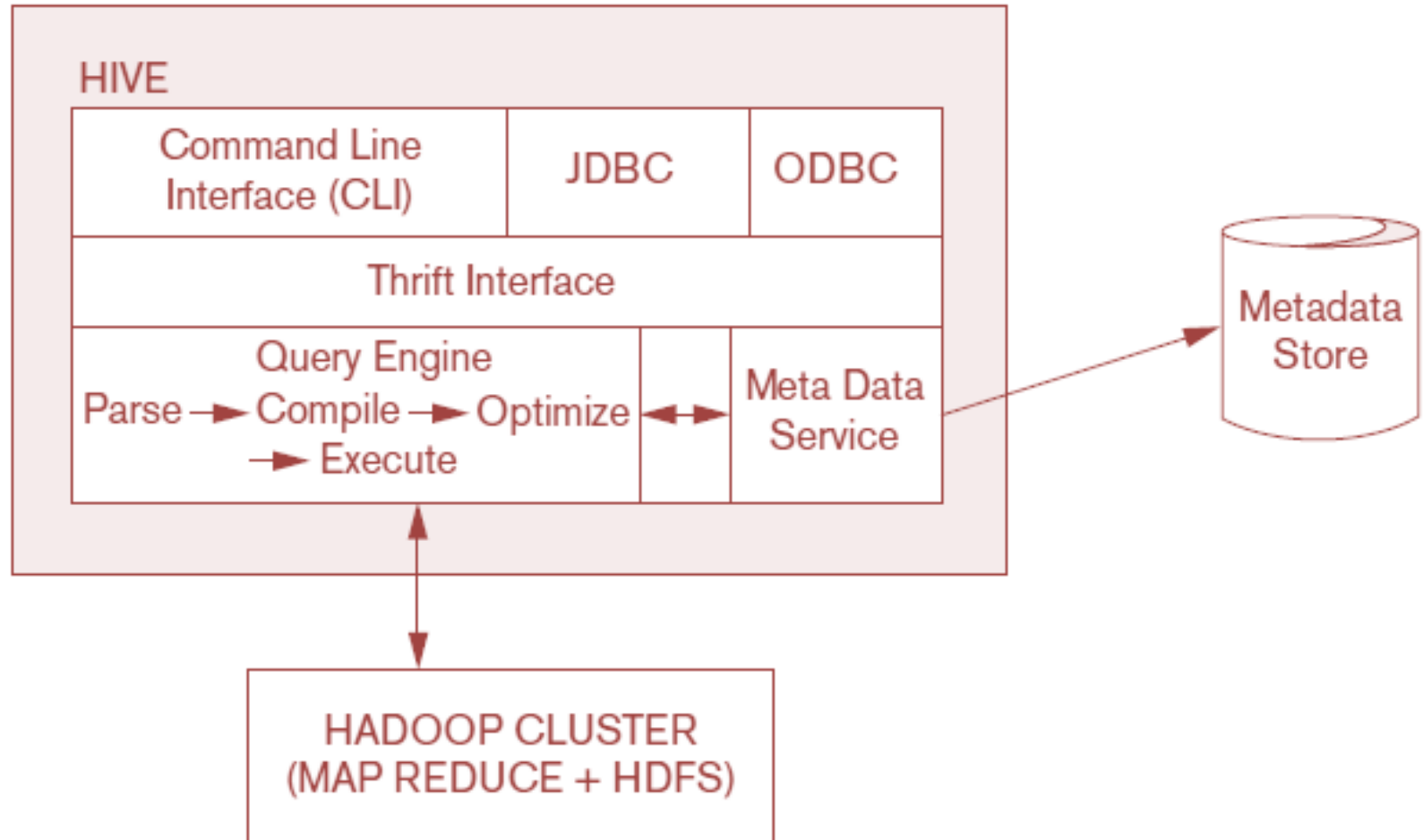


Figure: Hive system architecture and components

Advantages of the Hadoop/MapReduce Technology

- Disk seek rate a limiting factor when dealing with very large data sets
 - Limited by disk mechanical structure
- Transfer speed is an electronic feature and increasing steadily
- MapReduce processes large datasets in parallel
- MapReduce handles semi-structured data and key-value datasets more easily
- Linear scalability

6.5 Hadoop v2 (Alias YARN)

- Reasons for developing Hadoop v2
 - Job Tracker became a bottleneck
 - Cluster utilization less than desirable
 - Different types of applications did not fit into the MR model
 - Difficult to keep up with new open source versions of Hadoop

YARN Architecture

- Separates cluster resource management from Jobs management
- Resource Manager and Node Manager together form a platform for hosting any application on YARN
- Application Masters send Resource Requests to the Resource Manager which then responds with cluster Container leases
- Node Manager responsible for managing Containers on their nodes

Hadoop Version Schematics

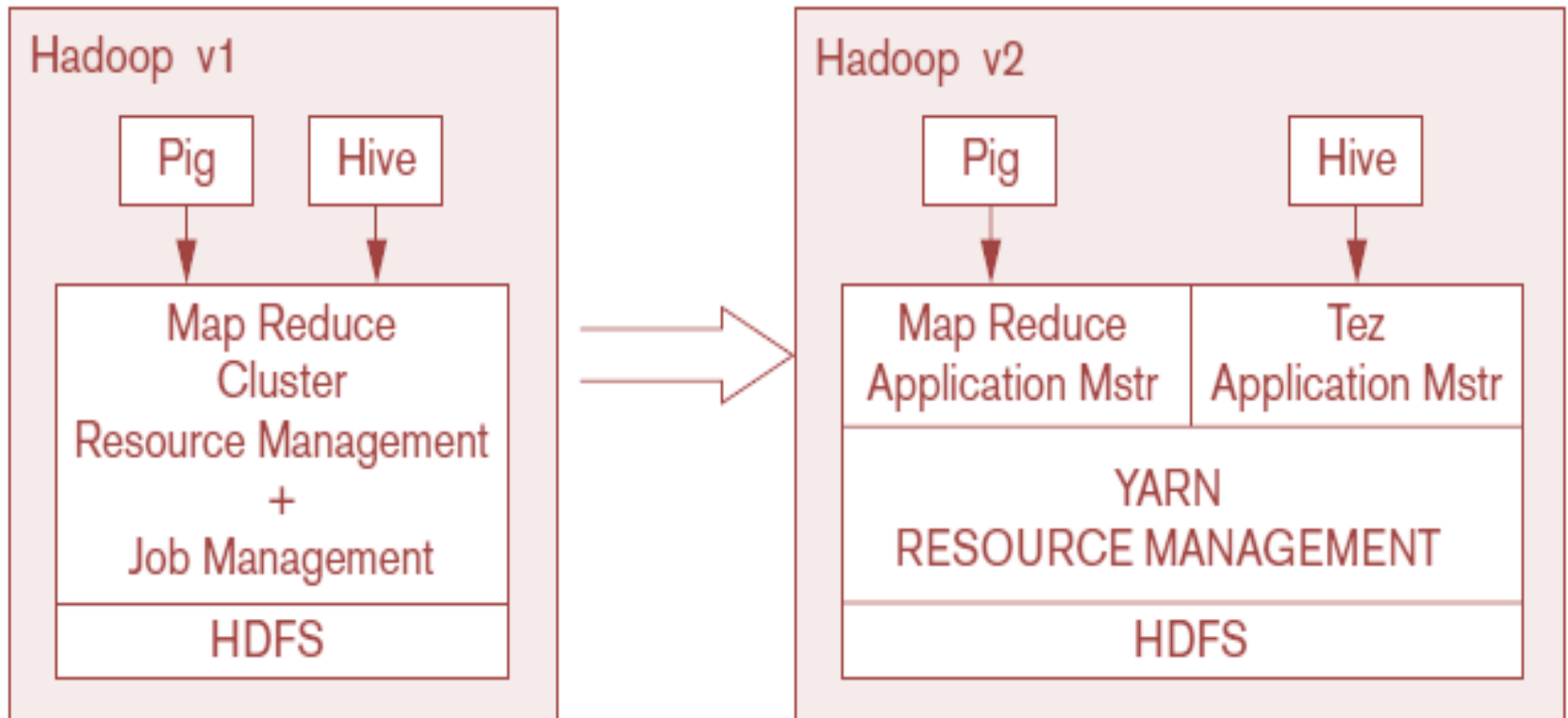


Figure: The Hadoop v1 vs. Hadoop v2 schematic

Other Frameworks on YARN

- Apache Tez
 - Extensible framework being developed at Hortonworks for building high-performance applications in YARN
- Apache Giraph
 - Open-source implementation of Google's Pregel system, a large-scale graph processing system used to calculate Page-Rank
- Hoya: HBase on YARN
 - More flexibility and improved cluster utilization

6.6 General Discussion

- Hadoop/MapReduce versus parallel RDBMS
 - 2009: performance of two approaches measured
 - Parallel database took longer to tune compared to MR
 - Performance of parallel database 3-6 times faster than MR
 - MR improvements since 2009
 - Hadoop has upfront cost advantage
 - Open source platform

Cont'd...

- MR able to handle semi-structured datasets
- Support for unstructured data on the rise in RDBMSs
- Higher level language support
 - SQL for RDBMSs
 - Hive has incorporated SQL features in HiveQL
- Fault-tolerance: advantage of MR-based systems

Cont'd...

- Big data somewhat dependent on cloud technology
- Cloud model offers flexibility
 - Scaling out and scaling up
 - Distributed software and interchangeable resources
 - Unpredictable computing needs not uncommon in big data projects
 - High availability and durability

Cont'd...

- Data locality issues
 - Network load a concern
 - Self-configurable, locality-based data and virtual machine management framework proposed
 - Enables access of data locally
 - Caching techniques also improve performance
- Resource optimization
 - Challenge: optimize globally across all jobs in the cloud rather than per-job resource optimizations

Cont'd...

- YARN as a data service platform
 - Emerging trend: Hadoop as a data lake
 - Contains significant portion of enterprise data
 - Processing happens
 - Support for SQL in Hadoop is improving
- Apache Storm
 - Distributed scalable streaming engine
 - Allows users to process real-time data feeds
- Storm on YARN and SAS on YARN

Cont'd...

- Challenges faced by big data technologies
 - Heterogeneity of information
 - Privacy and confidentiality
 - Need for visualization and better human interfaces
 - Inconsistent and incomplete information

Cont'd...

- Building data solutions on Hadoop
 - May involve assembling ETL (extract, transform, load) processing, machine learning, graph processing, and/or report creation
 - Programming models and metadata not unified
 - Analytics application developers must try to integrate services into coherent solution
- Cluster a vast resource of main memory and flash storage
 - In-memory data engines
 - Spark platform from Databricks

Thank You !

